

Machine Learning Principles

Class 19: November 13

Bayesian Net II (Gibbs Sampling & EM algorithm)

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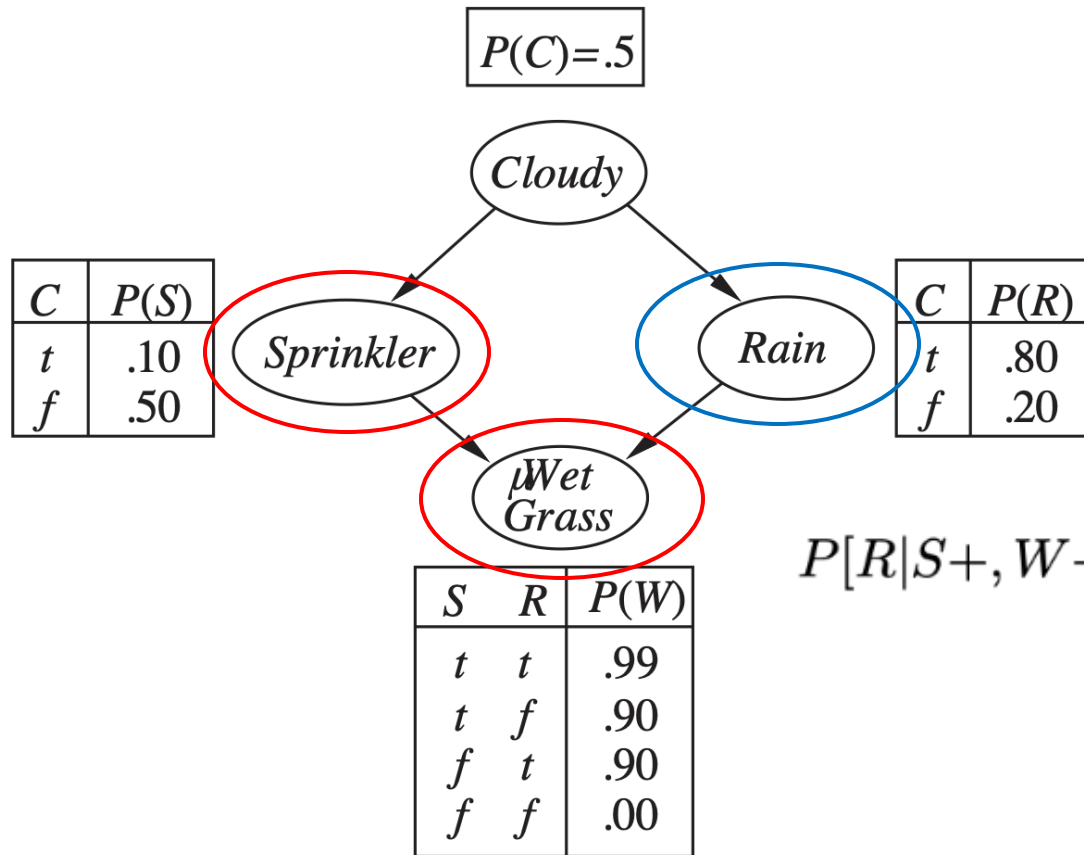
Today's Lecture

1. Inference: computing the posterior probability
 - Variable Elimination
 - Markov Chain Monte Carlo (MCMC) simulation (Gibbs Sampling)
2. Learning a Bayesian Net (learning parameters : CPT)
 - a network structure is known and data is fully observed
 - a network structure is known and data is partially observed
3. **E**xpectation **M**aximization (EM) algorithm (partial observation)
4. **G**aussian **M**ixture **M**odeling (GMM)

[1] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R \mid S: +, W: +]$?

- [Method 1: Variable Elimination]



$$\begin{aligned}
 P[R|S+, W+] &= \alpha \sum_C P[R, S+, W+, C] \\
 &= \alpha \sum_C P(C) \cdot P(S+|C) \cdot P(R|C) \cdot P(W+|S+R) \\
 &= \alpha P(W+|S+R) \sum_C P(C) \cdot P(S+|C) \cdot P(R|C)
 \end{aligned}$$

Figure 14.12 From the book “AI: A Modern Approach”

[2] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

■ [Method 2: Sampling]

$\#samples(Rain: +, Sprinkler: +, WetGrass: +)$

$\# samples (Sprinkler: +, WetGrass: +)$

# data	Rain	Cloudy	Sprinkler	WetGrass
1	+	+	+	+
2	-	-	+	+
3	+	+	+	+
4	+	-	+	+
5	-	+	+	+
6	-	-	+	+
...	-	-	+	+

Fig

[2] Inference (Markov Chain Monte Carlo Simulation)

Q: how to generate data samples, following the distribution?

[3] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

- (1) Gibbs sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**

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suppose the initial state is this. + fixed as evidence

[4] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

(1) sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**

suppose the initial state is this.

+ fixed as evidence

(2) Randomly sample a value for one of non evidence variable.
Rain or Cloud (in an arbitrary order)

[5] Inference (Markov Chain Monte Carlo Simulation)

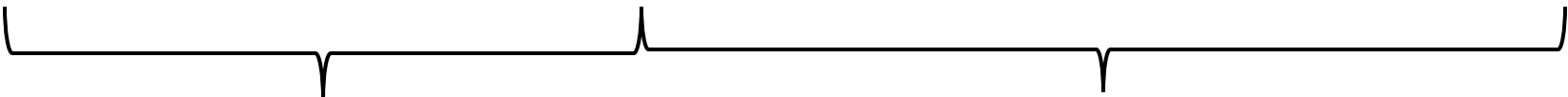
How to compute $P[C : + \mid R : +, S : +, W : +]$?
or how to compute $P[R : - \mid C : -, S : +, W : +]$?

[6] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R: +, W: +]$?

- (1) sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**


+ suppose the initial state is this. + fixed as evidence

- (2) Randomly sample a value for one of non evidence variable.
Rain or Cloud (in an arbitrary order)

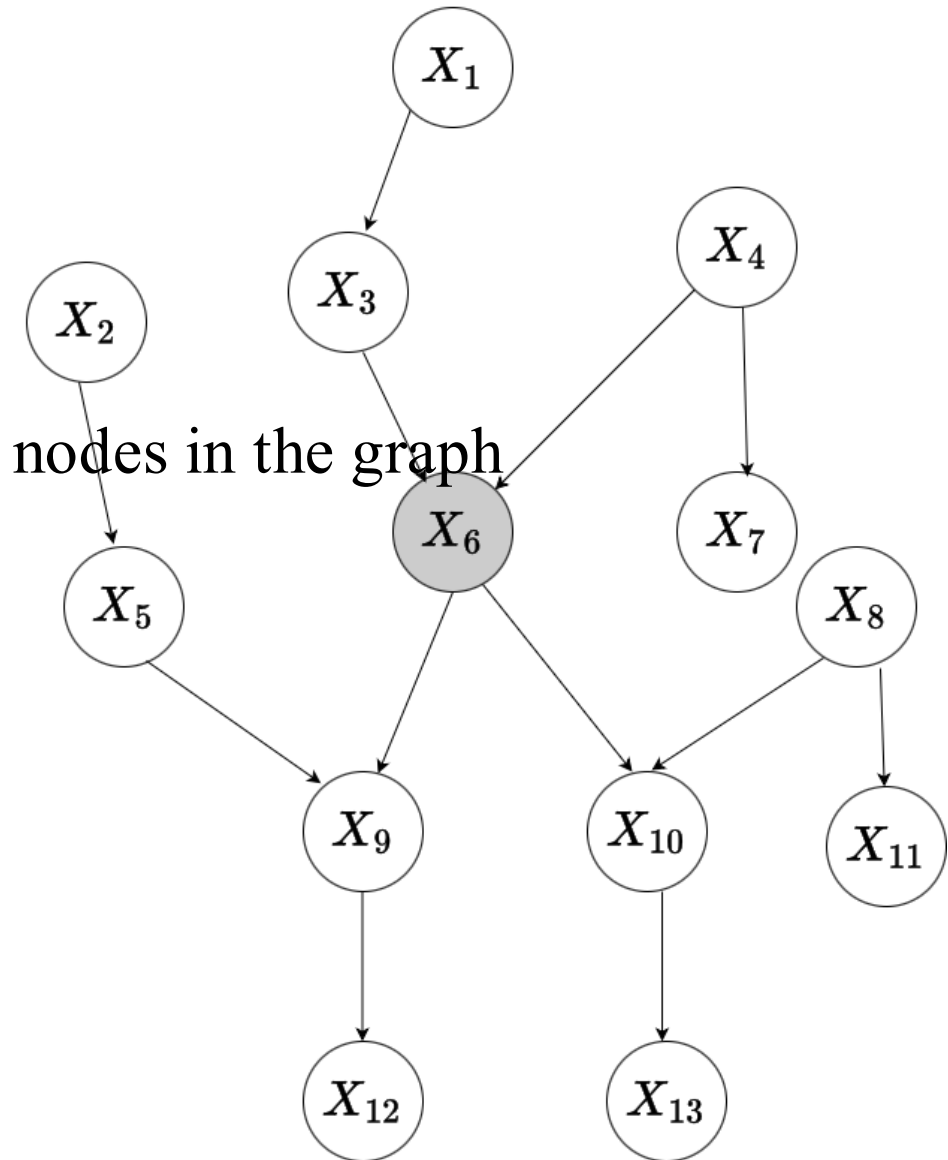
- (3) **Cloud is sampled by $P[\text{Cloud} | \text{Sprinkler: + and Rain: +}]$**
given the current value in the Markov Blanket Variables of “Cloud”.

Q: what is Markov Blanket?

** Markov Blanket

Markov Blanket of a Variable

The variable is conditionally independent of every other nodes in the graph given its Markov blanket.



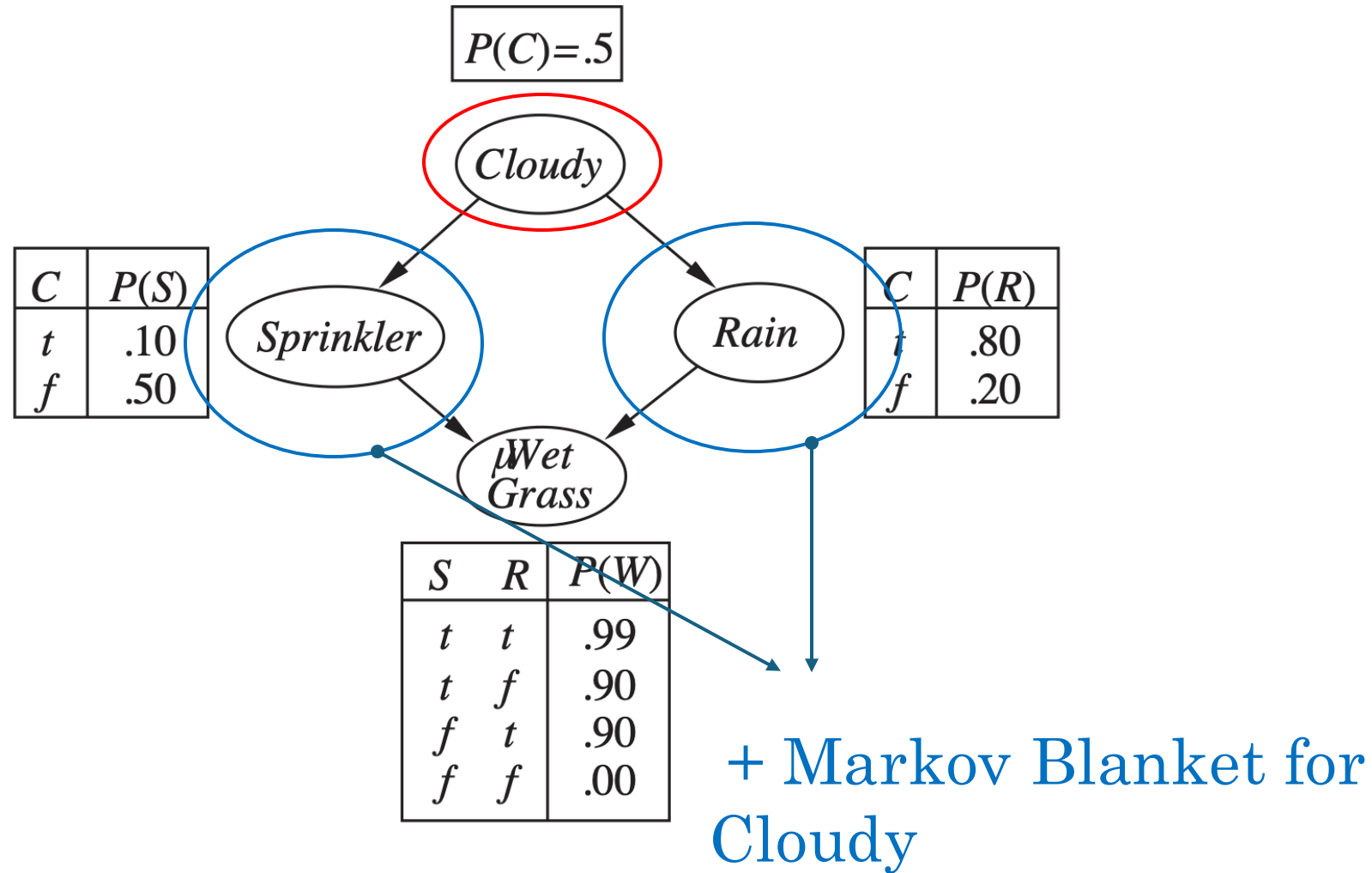
** Markov Blanket

**How to compute $P[\text{Target} \mid \text{Markov Blanket}]$?

Given Markov Blanket, the target variable is independent from other R.Vs, so we can imagine a new graph only with the target and blanket variables.

$$\begin{aligned} P[\text{Target} \mid \text{Markov Blanket}] &= \alpha P[\text{target}, \text{Markov Blanket}] \\ &= \alpha P[\text{target}, \text{parent}, \text{children}, \text{coparent}] \\ &= \alpha P[\text{parent}] P[\text{coparent}] P[\text{target} \mid \text{parent}] P[\text{children} \mid \text{coparents and target}] \\ &= \beta P[\text{target} \mid \text{parent}] P[\text{children} \mid \text{coparents and target}] \end{aligned}$$

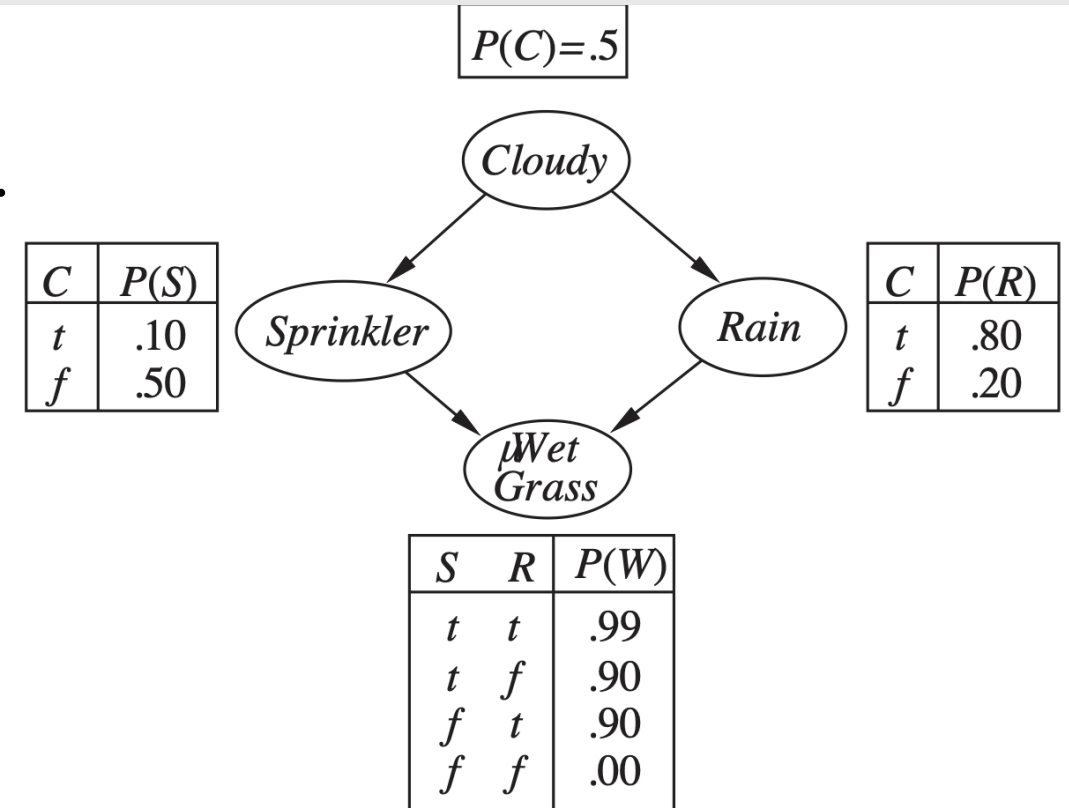
[6] Inference (Markov Chain Monte Carlo Simulation)



Q: What is the Markov Blanket for Rain?

[7] Inference (Markov Chain Monte Carlo Simulation)

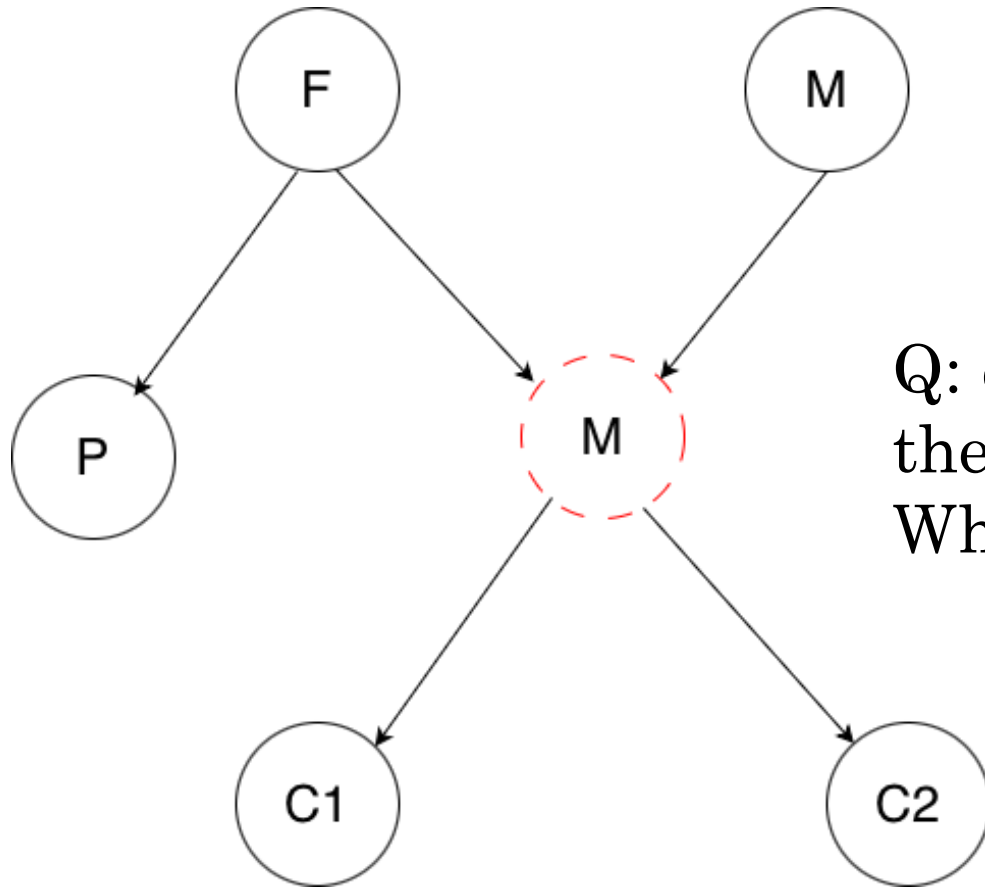
Example of Computing Posterior,
given Markov Blanket Variables as Evidence.



$P[\text{Cloud} \mid \text{Sprinkler: + \& Rain: +}]?$

$$\propto \frac{P[C +]P[\text{Sprinkler}+ \mid C+] P[\text{Rain}+ \mid C+]}{P[C -]P[\text{Sprinkler}+ \mid C-] P[\text{Rain}+ \mid C-]} = \propto \begin{bmatrix} 0.04 \\ 0.05 \end{bmatrix} \approx \begin{bmatrix} 0.44 \\ 0.56 \end{bmatrix}$$

[8] Inference (Application)



Q: could we measure the probability of that the person M is the family member ?
What do we need?

Learning a Bayesian Net

- we will focus on the case where a structure is given.

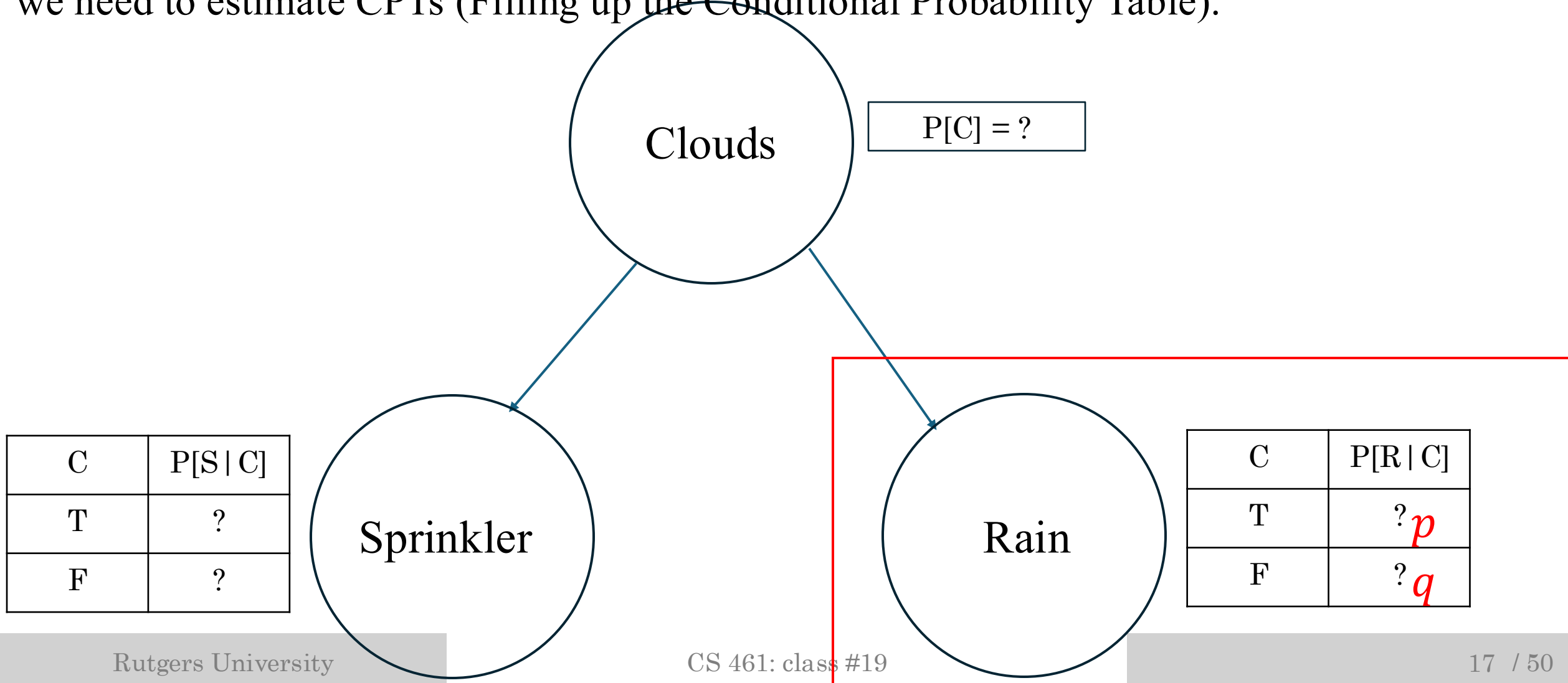
[1] Learning Bayesian Tree

If you are interested in learning a structure,
Chow-Liu Algorithm finds an optimal T (first / second order dependence)
 $T = \operatorname{argmin}_T KL(P || T)$, where P is a true distribution.

Q: what is the true distribution? How can we compute this?

[2] Learning Bayesian Tree (complete data observation)

When a structure is known and data is fully observed, we need to estimate CPTs (Filling up the Conditional Probability Table).



[3] Learning Bayesian Tree (complete data observation)

MLE estimation for p :
when data is fully observed.

$$p(D) = \prod_{n=1}^N P(x_n)$$

$$\log P(D) = \sum_{n=1}^N \log P(x_n)$$

$$= \sum_{n=1}^N \log P[C(x_n)] + \log P[S(x_n)|C(x_n)] + \log P[R(x_n)|C(x_n)]$$

$$\frac{\partial \log P(D)}{\partial p} = N_p \cdot 1/P - N_n \cdot 1/(1 - P) = 0$$

$$P = \frac{N_p}{N_p + N_n}$$

the four possible cases of R and C
: $p, 1 - p, q, 1 - q$

- N_p : # samples cloud + & rain +
- N_n : # samples cloud + & rain -

This is just a sample mean (Bernoulli R.V)

[4] Learning Bayesian Tree (partial data observation)

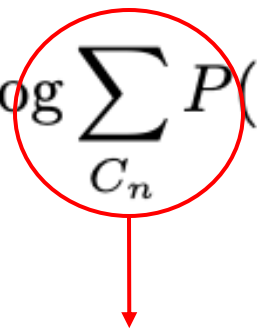
what if data is partially observed?

suppose we forgot to collect **cloud** information.

can we still estimate *p or q* : $P[\text{rain} | \text{cloud} +]$ or $P[\text{rain} | \text{cloud} -]$?

[5] Learning Bayesian Tree (partial data observation)

- MLE when data is partially observed.

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta, p, q)$$


marginalization over unobserved variables

- MLE when data is fully observed (complete MLE)

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log P(S_n, R_n, C_n|\theta, p, q)$$

Q: which one is easier in computing MLE ?

[6] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta, p, q) &= \sum_{n=1}^N \log P(S_n, R_n, C_n|\theta, p, q) \\ &= \sum_{n=1}^N \log P(C_n) + \log P(S_n|C_n) + \log P(R_n|C_n)\end{aligned}$$

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta, p, q)$$

$$\begin{aligned}&= \sum_{n=1}^N \log(P(C_n+) \cdot P(S_n|C_n+) \cdot P(R_n|C_n+) \\ &\quad + P(C_n-) \cdot P(S_n|C_n-) \cdot P(R_n|C_n-))\end{aligned}$$

hard to optimize (parameters are interdependent)!

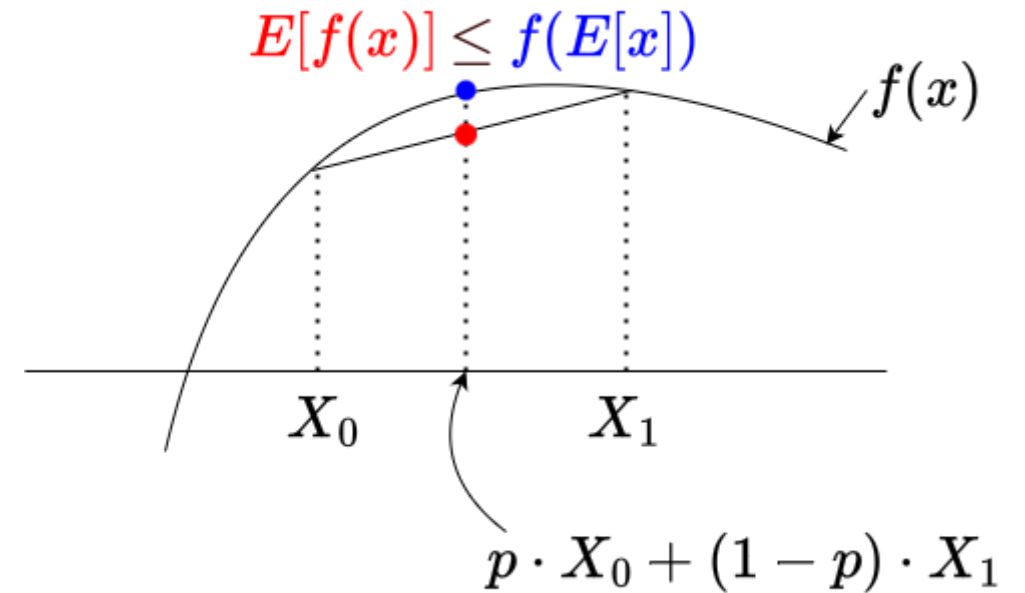
[7] Learning Bayesian Tree (partial data observation)

It's hard to get a closed form to compute MLE,
when there are missing variables.

($\log \sum$ is not tractable for MLE; can we change it to $\sum \log$ like the complete case?)

[8] Learning Bayesian Tree (partial data observation)

Jensen's Inequality: (both expectation are finite)
 $E[\log X] \leq \log E[X]$



[9] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta) &= \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta) \\ &= \sum_{n=1}^N \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta)q(C_n)}{q(C_n)} \\ &= \sum_{n=1}^N \log E\left[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right] \\ &\geq \sum_{n=1}^N E\left[\log\left(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right)\right] \\ &\geq \sum_{n=1}^N \underbrace{E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))}\end{aligned}$$

- hard to optimize!

- by Jensen's Inequality

- easy to optimize!

expectation over

$q(C_n)$: any arbitrary density for C_n

gives a lower bound.

Q: how can we make the lower bound tight?

[10] Learning Bayesian Tree (partial data observation)

Q: What does the expectation of likelihood mean?

Likelihood is not a probability. It is a function of the parameters.

In the example below, the likelihood with one point data x_i is a function of $L(\mu, \sigma)$.

$$\mathcal{N}(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\left\{\frac{-1}{2\sigma^2}(x_i - \mu)^2\right\}$$

What if x_i is unobserved and instead we know the density $x_i \sim p(x_i)$?
we can measure the expectation treating μ *and* σ as constants.

$$E[f(X_i|\mu, \sigma)] = \sum_{x_i} f(x_i|\mu, \sigma)p(x_i)$$

[11] Learning Bayesian Tree (partial data observation)

Q: what density makes the lower bound tight?

$$\begin{aligned} & \sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta) + H(q(C_n))] \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(S_n, R_n, C_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(C_n | S_n, R_n, \theta) P(S_n, R_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \left(\sum_{C_n} q(C_n) \log \frac{P(C_n | S_n, R_n, \theta)}{q(C_n)} + \sum_{C_n} q(C_n) \log P(S_n, R_n | \theta) \right) \\ &= \sum_{n=1}^N -KL(q(C_n) || P(C_n | S_n, R_n, \theta)) + \log P(S_n, R_n | \theta) \end{aligned}$$

when $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

the lower bound becomes equal to $\log P(S_i, R_i | \theta, p, q)$

[12] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta) &= \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta) \\ &= \sum_{n=1}^N \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta)q(C_n)}{q(C_n)} \\ &= \sum_{n=1}^N \log E\left[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right] \\ &\geq \sum_{n=1}^N E\left[\log\left(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right)\right] \\ &\geq \sum_{n=1}^N E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))\end{aligned}$$

expectation over $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

what is the next ?

we do maximization over θ, p, q . why?

Expectation Maximization:
finding maximum likelihood when data is partially observed.

[1] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q
2. compute $P(C_n | S_n, R_n, \theta, p, q)$ for $\forall n$
3. **E step**: compute

$$\sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta, p, q)]$$

4. **M step**: Update $\theta, p, q = \operatorname{argmax}_{\theta, p, q} \sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta, p, q)]$
5. go back to step 2.

[2] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q

2. compute $P(C_i | S_i, R_i, \theta, p, q)$ for $\forall i$

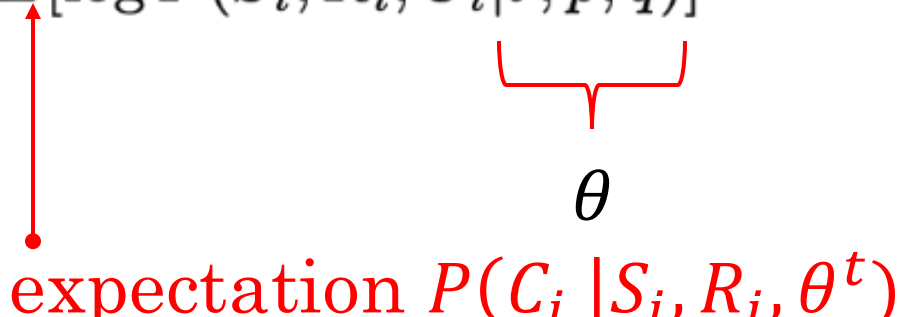
3. E step: compute $\sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)] = Q(\theta, \theta^t)$

$\downarrow$
Auxiliary Function

4. S step: Update $\theta, p, q = \operatorname{argmax}_{\theta, p, q} \sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)]$

5. go back to step 2.

$$\text{Auxiliary Function } Q(\theta, \theta^t) = \sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)]$$



- $Q(\theta^t, \theta^t) = \sum_i \log P[S_i, R_i | \theta^t]$
- $Q(\theta^{t+1}, \theta^t) \geq Q(\theta^t, \theta^t)$ by **Maximization** EM monotonically increases the observed data log likelihood!
- $Q(\theta^{t+1}, \theta^{t+1}) \geq Q(\theta^{t+1}, \theta^t)$ by **lower bound**
- $Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$ by **Expectation**
- $\sum_i \log P[S_i, R_i | \theta^t] \leq Q(\theta^{t+1}, \theta^t) \leq Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$

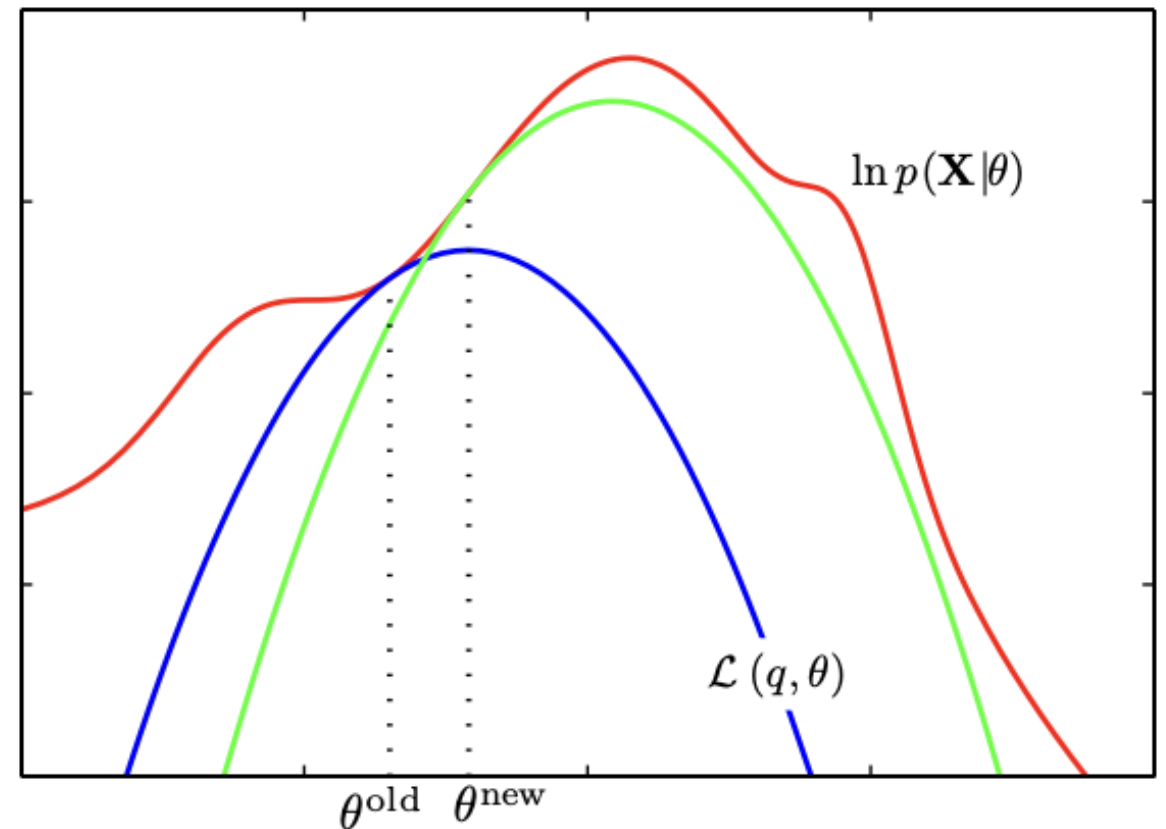
[3] EM algorithm (original likelihood vs. auxiliary function)

EM algorithm finds a local minimum.

$\log P(\mathbf{X}|\theta)$ is often non – convex or non – concave.

Figure 9.14 The EM algorithm involves alternately computing a lower bound on the log likelihood for the current parameter values and then maximizing this bound to obtain the new parameter values. See the text for a full discussion.

From textbook Bishop



- Comparison between EM and gradient based methods

Gaussian Mixture Model (GMM)

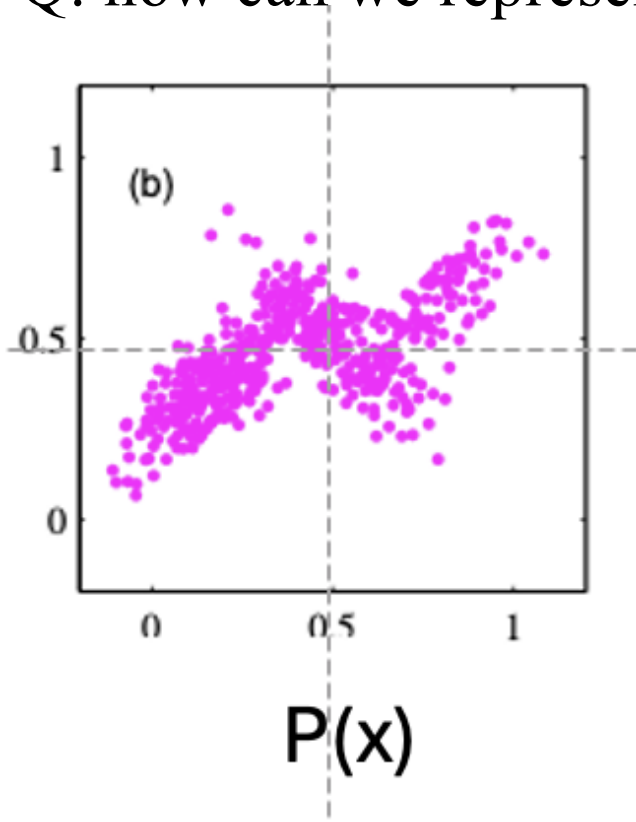
- density modeling (learning a Bayesian net)
 - what R.Vs (invisible & visible)
 - how they are connected
 - CPTs

[1] GMM (motivation)

Data often has multiple modalities like the figure below.

Q: continuous / discrete? How many dimensions?

Q: how can we represent the density $f(x_1, x_2)$: ?

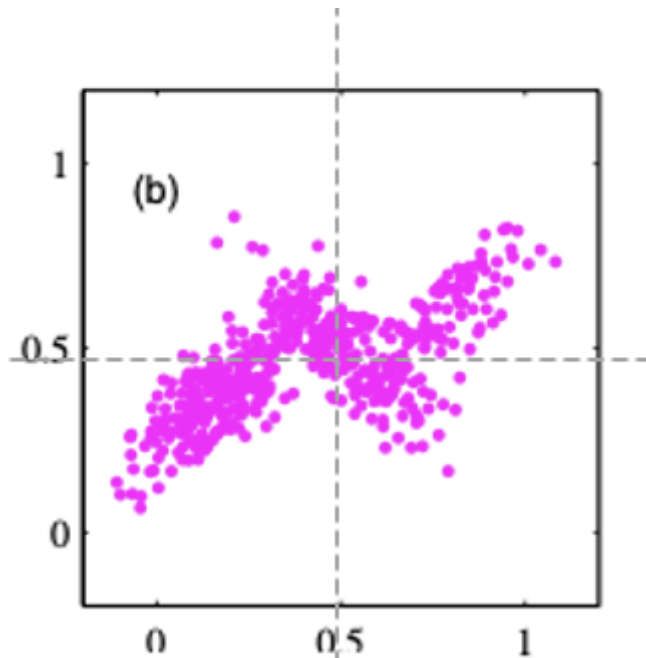


from textbook Bishop Fig. 9.5

- density modeling $P(X)$
 - what R.Vs
 - how they are connected
 - CPTs

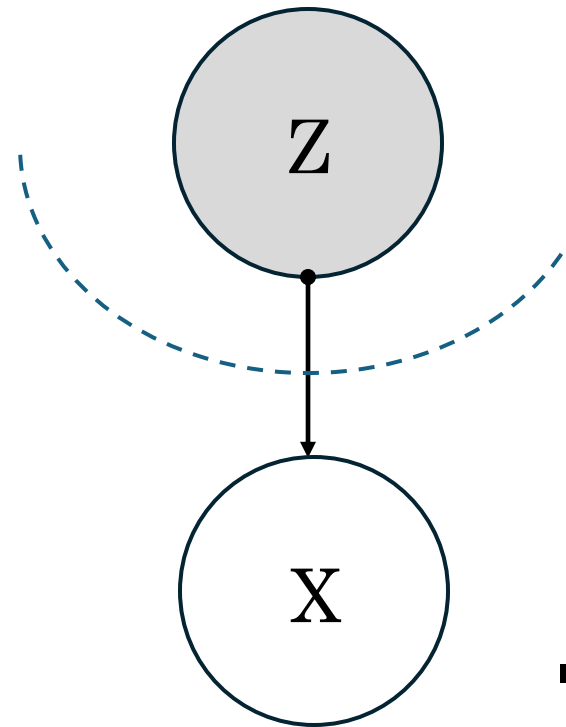
[2] GMM (motivation)

Q: what if we introduce a new R.V but invisible to us?



$P(x)$

from textbook Bishop Fig. 9.5

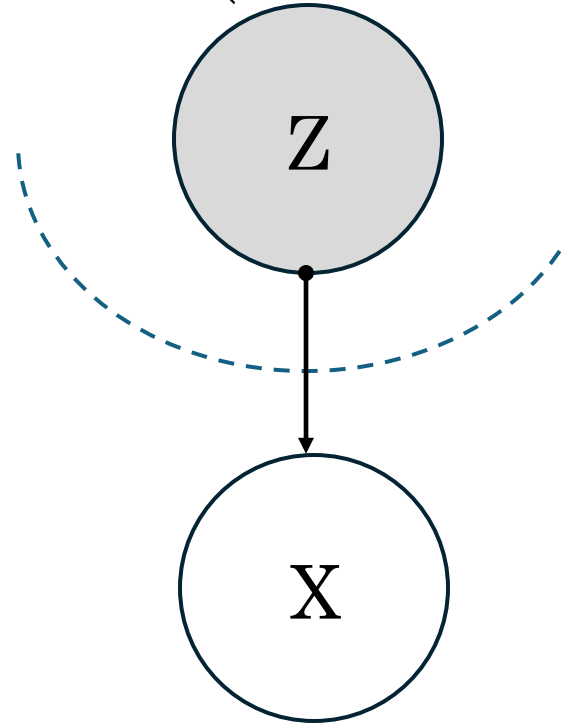


[with Z]

- density modeling $P(X)$
 - what R.Vs
 - how they are connected
 - CPTs

[3] GMM (hierarchical structure with a latent variable)

inaccessible (invisible / latent)

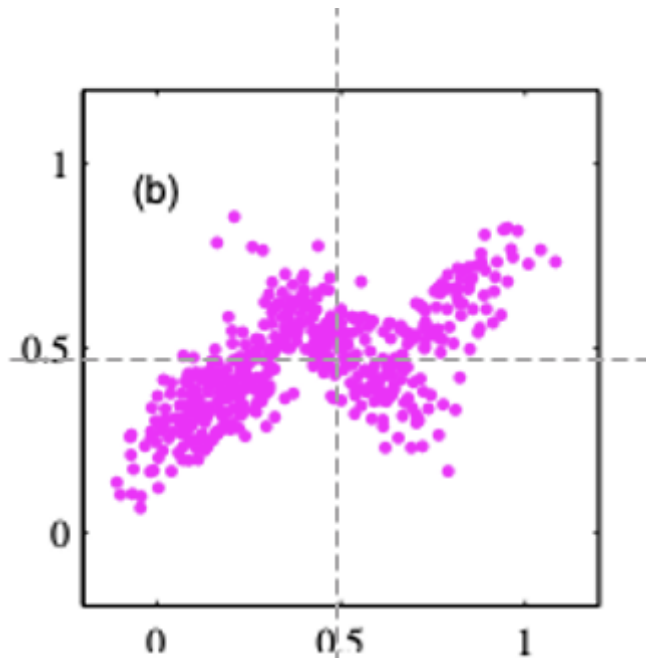


- by introducing the latent variable **Z**, the complex density (multi modal) can be decomposed into the simple densities (k-ary discrete & Gaussian)

[4] GMM (hierarchical structure with a latent variable)

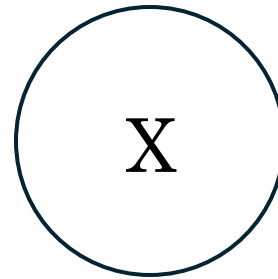
which model is structured so efficient and scalable?

Is the second model, Z accessible?



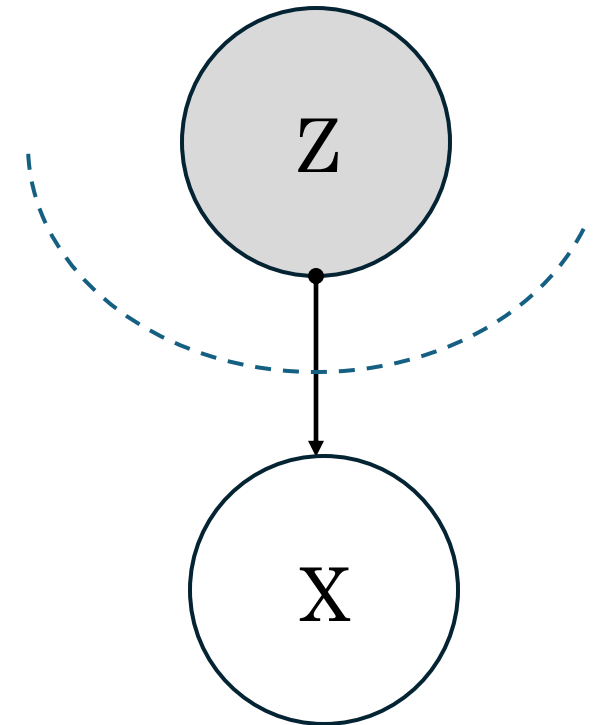
$P(x)$

from textbook Bishop Fig. 9.5



[without Z]

vs.



[with Z]

[5] GMM

Q: how can we generate Gaussian Mixture (GMM) samples?

[6] GMM Q: how can we generate Gaussian Mixture (GMM) samples?

[Generate N GMM samples]

- suppose $Z \sim K$ -ary discrete R.V
- $P(X|Z = k) \sim \text{Gaussian R.V} \sim N(\mu_k, \Sigma_k)$

[7] GMM Q: how can we generate Gaussian Mixture (GMM) samples?

[Generate N GMM samples]

- suppose $Z \sim K$ -ary discrete R.V
- $P(X|Z = k) \sim \text{Gaussian R.V} \sim N(\mu_k, \Sigma_k)$

(1) generate the samples for $Z \in \{0, 1, 2\}$.

(2) depending on z ,

choose a Gaussian density and generate a sample x according to the Gaussian

(3) repeat (1) and (2) for N times

[8] GMM (generation example)

(1) generate the samples for $\mathbf{z} \in \{0, 1, 2\}$.

$$P(\mathbf{z} = 0): 0.3, P(\mathbf{z} = 1): 0.2, P(\mathbf{z} = 2): 0.5$$

(2) depending on $\mathbf{z}$, choose the a Gaussian density.

$$\mathbf{G}_0: \left(\begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \right)$$

$$\mathbf{G}_1: \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \right)$$

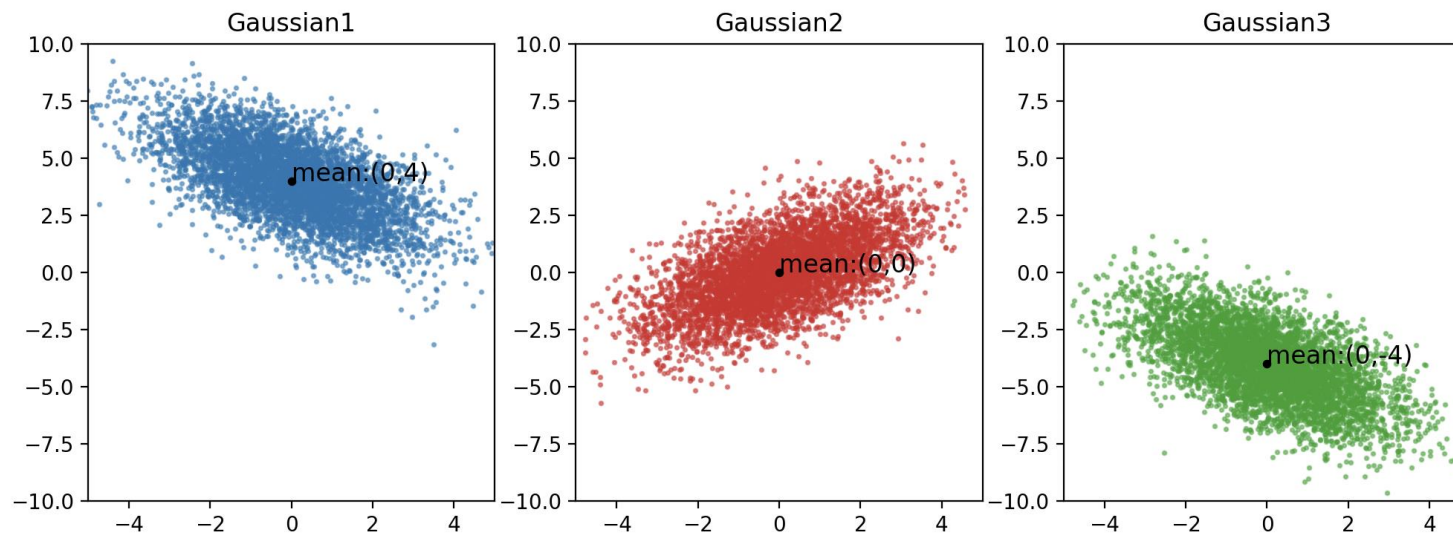
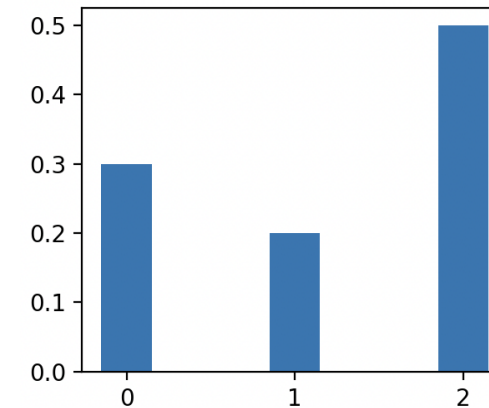
$$\mathbf{G}_2: \left(\begin{bmatrix} 0 \\ -4 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \right)$$

[9] GMM (generation example)

How can we generate Gaussian Mixture Modeling (GMM) samples?

(1) generate the samples for $Z \in \{0, 1, 2\}$.

(2) depending on Z , choose the a Gaussian density.

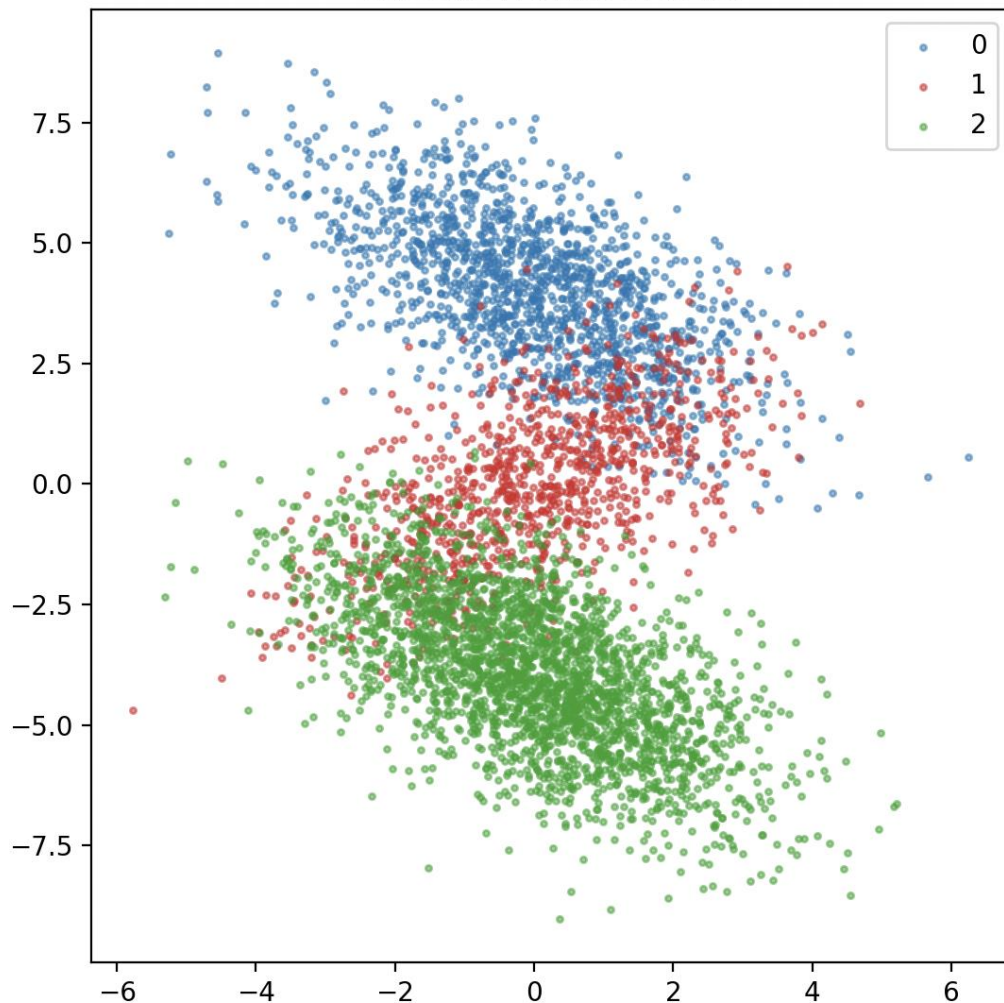


[10] GMM (generation example)

Q: Can you imagine the collective GMM samples (10,000)?

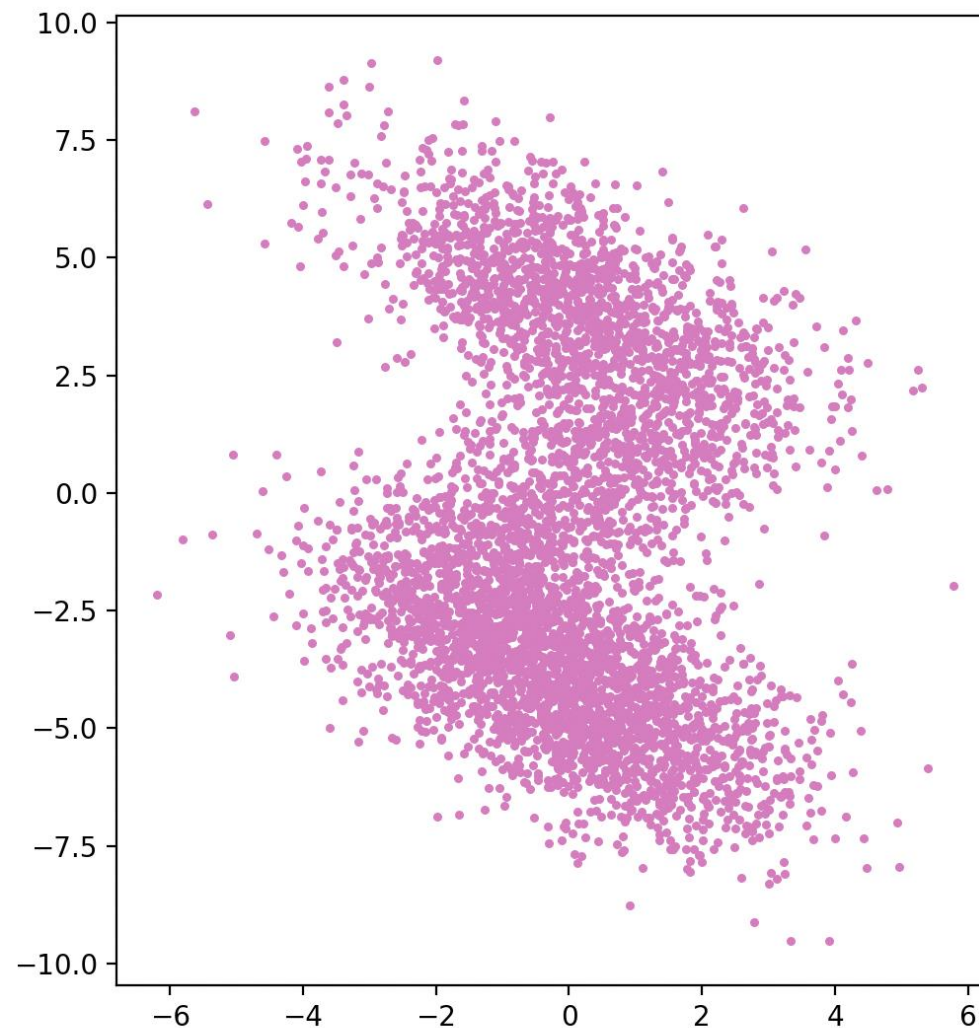
[11] GMM (example)

Gaussian Mixture Model



[when z is known]

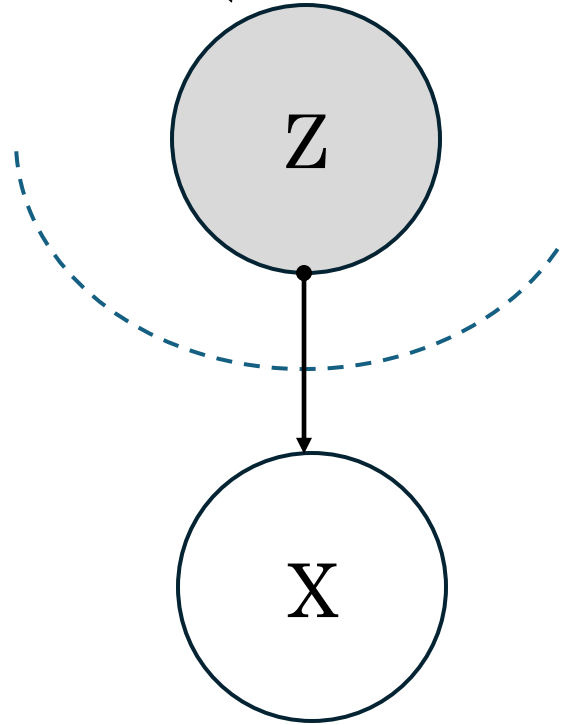
Gaussian Mixture Model



[when z is unknown]

[12] GMM (parameters)

inaccessible (invisible / latent)

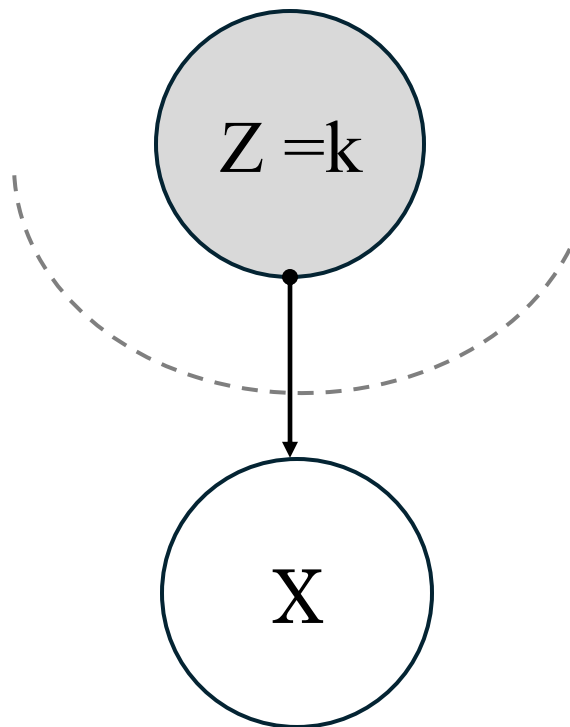


Q: what parameters do we need to define the model?

Q: how could we learn the parameters from data?

[13] GMM (parameters)

- Bayesian Network Representation of GMM



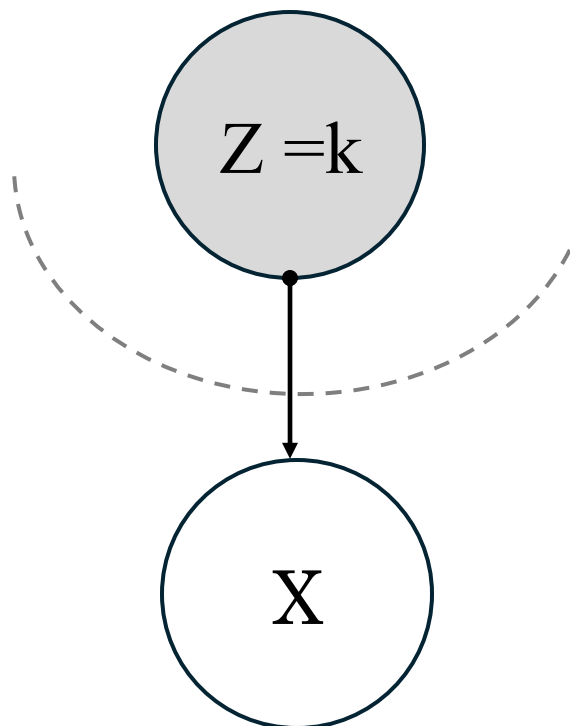
- $P(Z = k) = \pi_k$

- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

Q: $P[X]$?

[14] GMM (parameters)

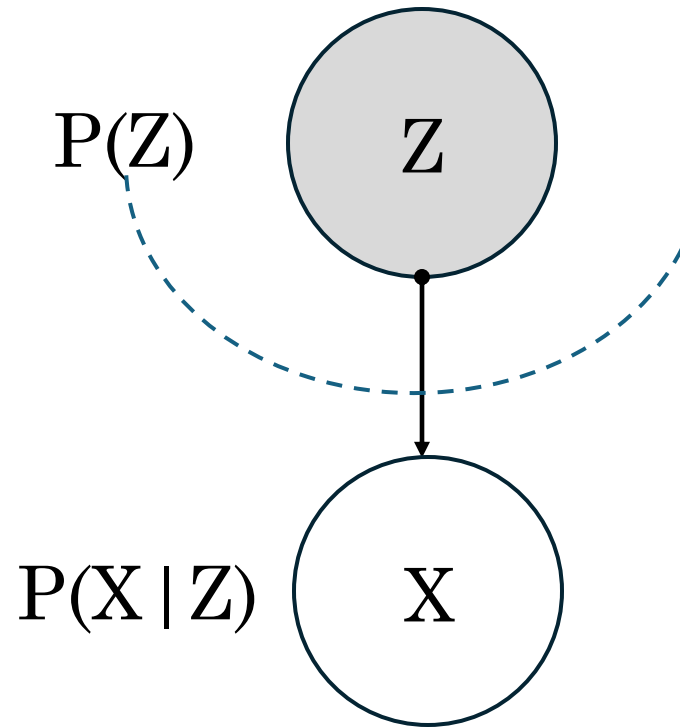
- Bayesian Network Representation of GMM



- $P(Z = k) = \pi_k$

- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

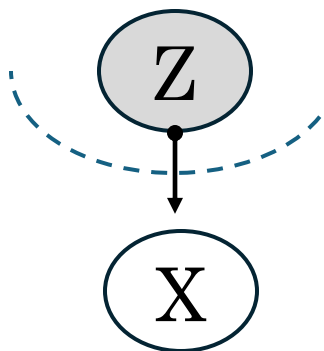
$$\begin{aligned} P(X) &= \sum_{Z=k} P(X, Z) = \sum_{Z=k} P(Z = k)P(X|Z = k) \\ &= \sum_{Z=k} \pi_k \mathcal{N}(\mu_k, \Sigma_k) \end{aligned}$$



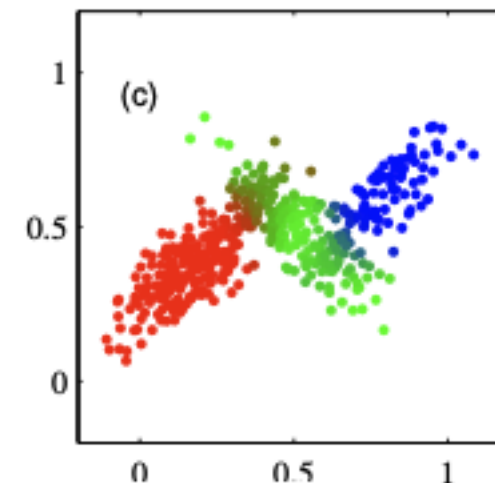
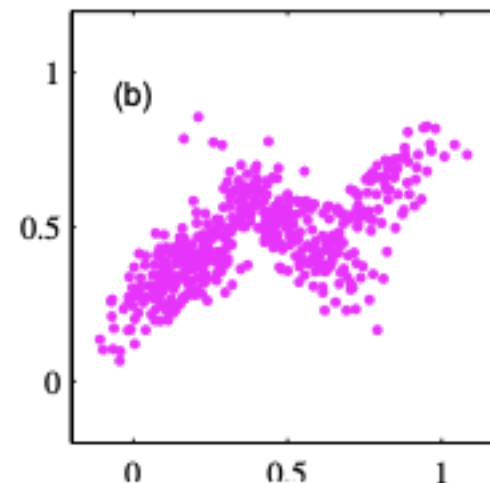
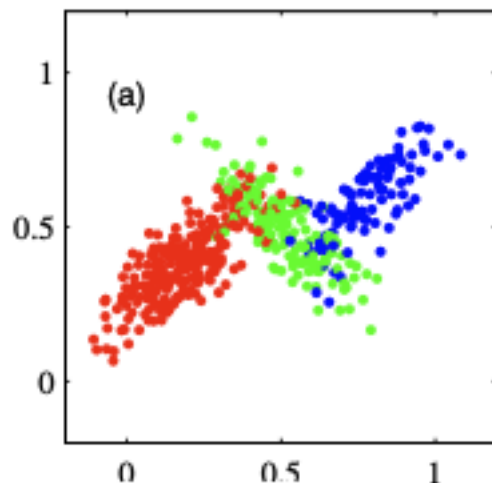
once we set this model,
we could compute the followings.

- $P(X)$
- $P(Z = k | x)$
 - * valuable information
 - * prediction about invisible factors.
 - * useful for clustering (when z is discrete)

[16] GMM (example of representation learning)



$$P(Z, X) = P(Z) \cdot P(X|Z)$$



$P(Z|x)$

- Representation will be $Z = \operatorname{argmax} P(Z|x)$
- the representation will be one of colors: red / green / blue.

from textbook Bishop Fig. 9.5

Q: Learning a GMM:

is to learn the parameters : π, μ, Σ (structure is given)
but we cannot access Z (invisible).

how can we learn the parameters?
what method can we use?